# Protein–Protein Interactions
## using Scipion-chem

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# Protein–protein interactions

Protein–protein interactions are important in molecular and cellular biology for catalysis, regulation, immune response, and inhibition. Therefore, it is of interest to decipher PPI for the understanding of molecular biology events.

*Experimental methods*
Such as yeast two-hybrid and biophysical (cryo-EM, X-ray crystallography and nuclear magnetic resonance) methods have been used for the study of PPI.

However, these methods are usually both tedious and time-consuming for large-scale and routine application.

**Hence, it is of interest to develop computer-aided tools for the analysis and prediction of PPI.**

# Protein–protein interactions

Many disease-associated variants sit at the interface. They change binding affinity or specificity.

## Classic examples

- hormone–receptor
- protease–inhibitor
- antibody–antigen

## What we need for *in silico* studies

**High-resolution structures** of protein–protein complexes give the chemical basis for understanding PPIs at the atomic level.

# Hot spots at the interface

Although protein–protein interfaces usually involve a large number of residues, only a few residues in the protein–protein interface contribute to the stability of binding.

**Hot spots**

***Experimentally***

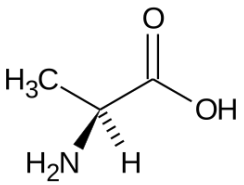
The identification of hot spots is performed by **alanine scanning** in which a specific residue is mutated to alanine and the difference in binding free energy before and after the mutation is measured to determine whether the specific residue is a hot spot or not.

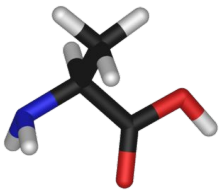
**Alanine Scanning**

- Typically, a binding energy difference of **2 kcal/mol** is used as the criterion for being a hot spot.

**2 kcal/mol**

**Hot spot**
ΔΔG > 0 — the mutation weakens the complex.
The residue was holding the interface together.



# The opposite question: improving affinity

- So, in contrast to hot spots, we have an affinity improver (negative ΔΔG value).
- On the other hand, we want to discover the residues that can improve the affinity and binding.
- In this case, we mutate to **any natural amino acid**.

**Affinity improver**
ΔΔG < 0 — the mutation strengthens the complex
This is what we are looking for in the Scipion workflow.

However, since *experimental mutagenesis* is expensive and time-consuming, **computational alanine scanning and mutagenesis** are highly desired to predict hot spots and affinity improvers.

# Protein–protein binding free energy

## Free energy

$$\Delta G = RT \ln (K_D)$$

## Binding free energy

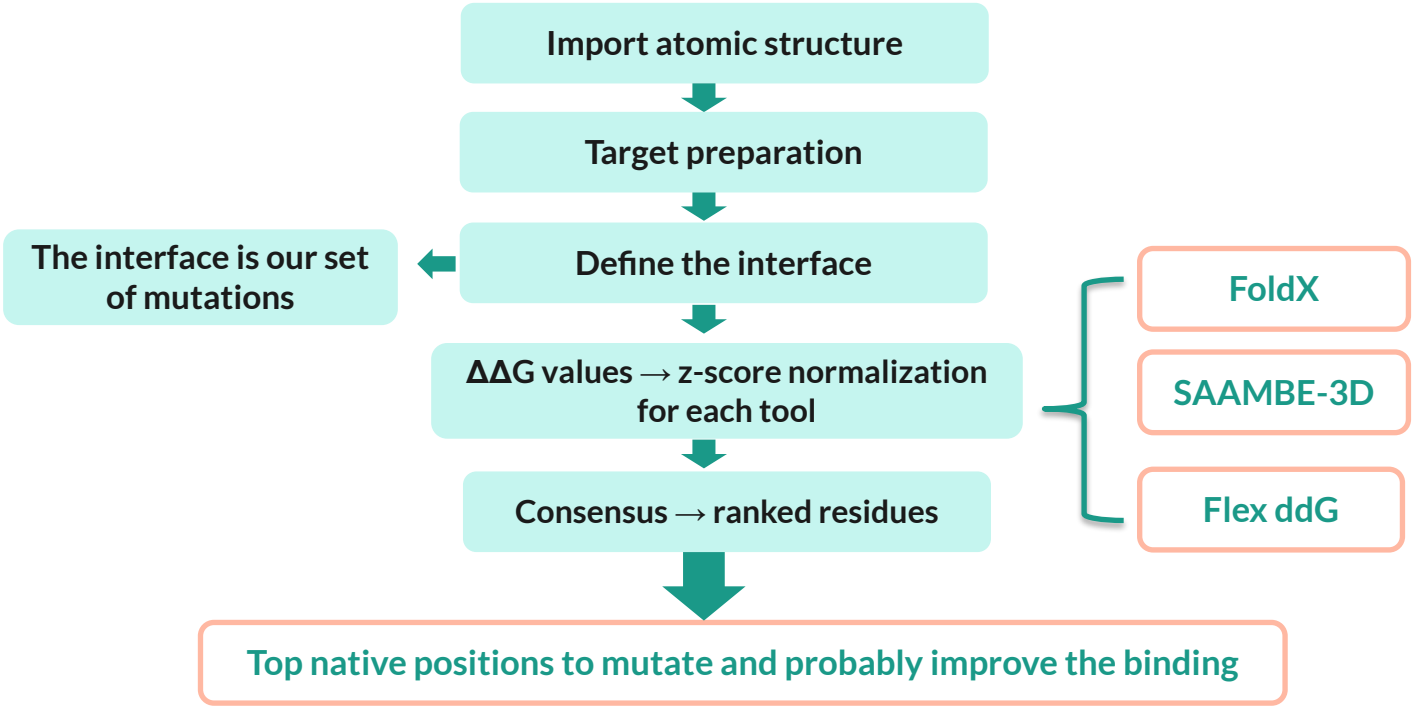
$$\Delta\Delta G_{bind} = \Delta G_{mutant} - \Delta G_{wild\text{-}type} \quad \text{kcal mol}^{-1}$$

- The binding free energy follows from the basic definition of free energy in protein–protein binding, expressed by the statistical definition.
- The difference in the calculated free energies (**ΔΔG**) between the final state (the mutant) and the reference state (the wild-type protein).
- We obtain **one ΔΔG per mutation**; that single number is what every tool predicts.

**Our aim: predict the change of binding free energy**

| | | |
|---|---|---|
| ΔΔG < 0 | Better affinity | Favorable mutation |
| ΔΔG > 0 | Worse affinity | Unfavorable mutation |

# Workflow in Scipion-chem



## Define the interface

The interface defines which residues are worth mutating, so the criterion matters.

### ROI

- Distance between atoms
- All-vs-all atom distance between the two chains
- Any atom pair ≤ one cutoff
- **3 Å cutoff.** The manual ROI default is strict; the literature usually uses 4–5 Å.
- Interaction type: no

### COCaDA

- Contact detection by atom chemistry
- Alpha-carbon (Cα) distance matrices, intra- and inter-chain contacts
- Classifies them into seven types: hydrogen and disulfide bonds; hydrophobic effects; attractive, repulsive, and salt-bridge interactions; aromatic stackings
- Interaction type: yes

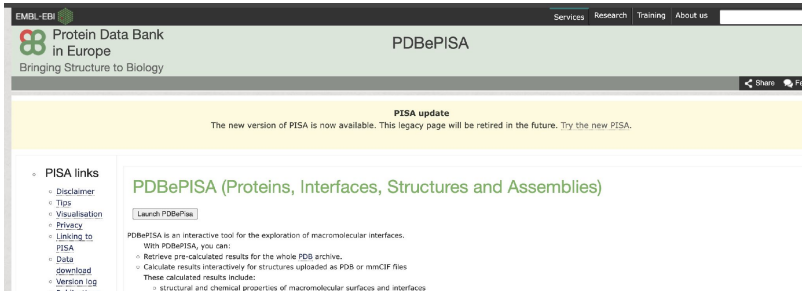
doi.org/10.3389/fbinf.2025.1630078
github.com/LBS-UFMG/COCaDA

# Define the interface

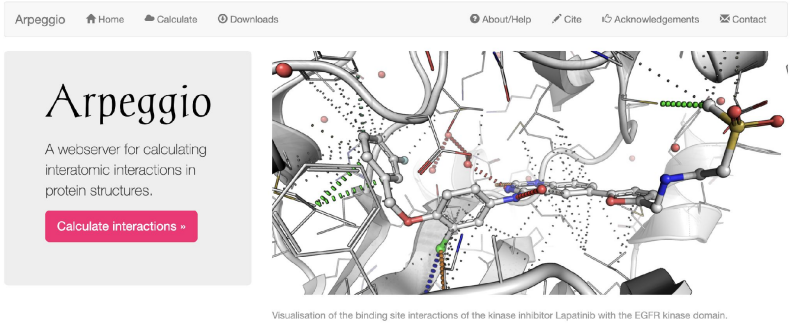
Do you know another software or tool to calculate it?

## Outside Scipion-chem

### PDBePISA




### Arpeggio



# Normalization: from ΔΔG to z-score

**ΔΔG values → z-score normalization**

Every tool has its own scale and its own systematic bias, so raw ΔΔG values cannot be compared between tools. We normalize each tool separately over the whole set of mutations.

$$z = (\Delta\Delta G - \mu) / \sigma$$

- μ : mean ΔΔG of the set
- σ : standard deviation of the set
- computed per tool, not across tools

| z-score | Reading | Candidate |
|---|---|---|
| $z > +1.5$ | Strongly destabilizing | Hot spot |
| $-1.5 \leq z \leq +1.5$ | Within the bulk | No clear effect |
| $z < -1.5$ | Strongly stabilizing | Affinity improver |

**The z-score is a position inside your set, not an energy.**
Change the set of mutations and μ, σ, and the z-scores change with it.

Remember: μ and σ come from your own set of mutations.

# Three ways of predicting the same number

Same input, same ΔΔG, three different routes to it:

| | FoldX | SAAMBE-3D | Flex ddG |
|---|---|---|---|
| **Approach** | Empirical force field | Machine learning (XGBoost) | Rosetta backrub ensemble |
| **What moves** | Side chains only | Nothing is modelled | Backbone and side chains |
| **Built on** | >1000 mutations | SKEMPI v2.0 (database) | ZEMu benchmark set |
| **Time per mutation** | Seconds | 0.2 s | 1–2 h on one core |

foldxsuite.crg.eu · SAAMBE-3D: doi 10.3390/ijms21072563 · Flex ddG: doi 10.1021/acs.jpcb.7b11367

## Consensus → ranked residues

### Why

- **Different assumptions:** a force field, a machine-learning model and an ensemble simulation.
- **Different scales:** the z-score is what makes them comparable.
- **Different failures:** the tools get different residues wrong, not the same ones.
- **Agreement between independent methods** is stronger evidence than a large value from one of them.

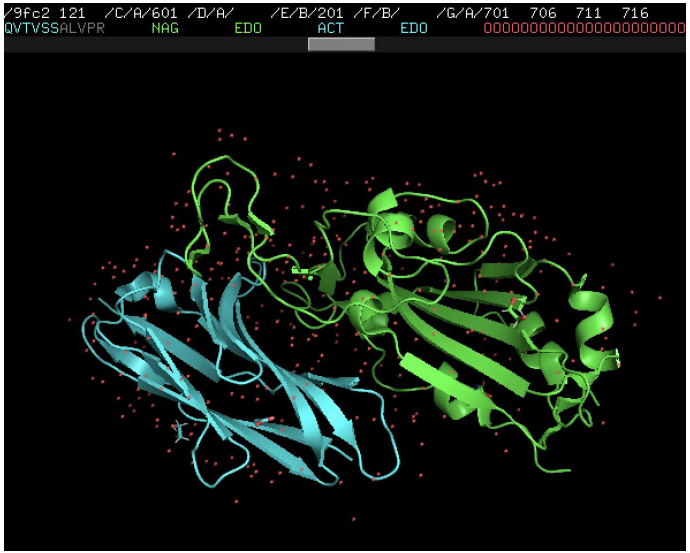
### How

1. Run the same mutations with the three tools
2. z-score each tool separately
3. Label it: $z > +1.5$, $z < -1.5$, or neutral
4. Count how many tools agree

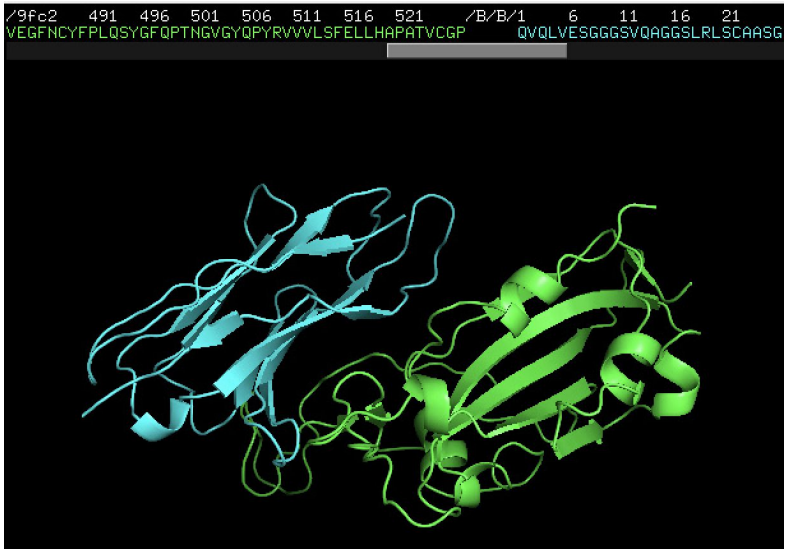
| | |
|---|---|
| 3 / 3 agree | Shortlist for experimental validation |
| 2 / 3 agree | Check the structure and the outlier |
| Signs conflict | Flexible or buried position — revisit |

# Visualize the output

**Import atomic structure**



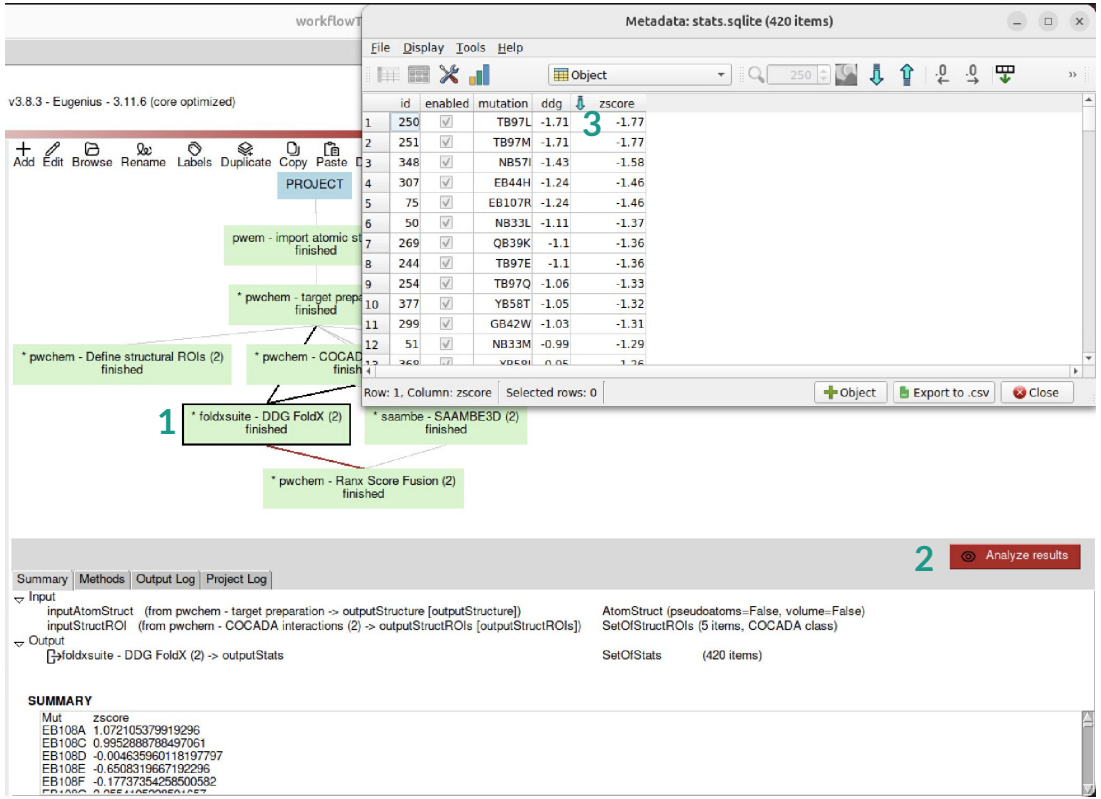

**Target preparation**




Same starting point for the three tools: one prepared complex, one interface, one set of mutations.

# Rank mutations by z-score

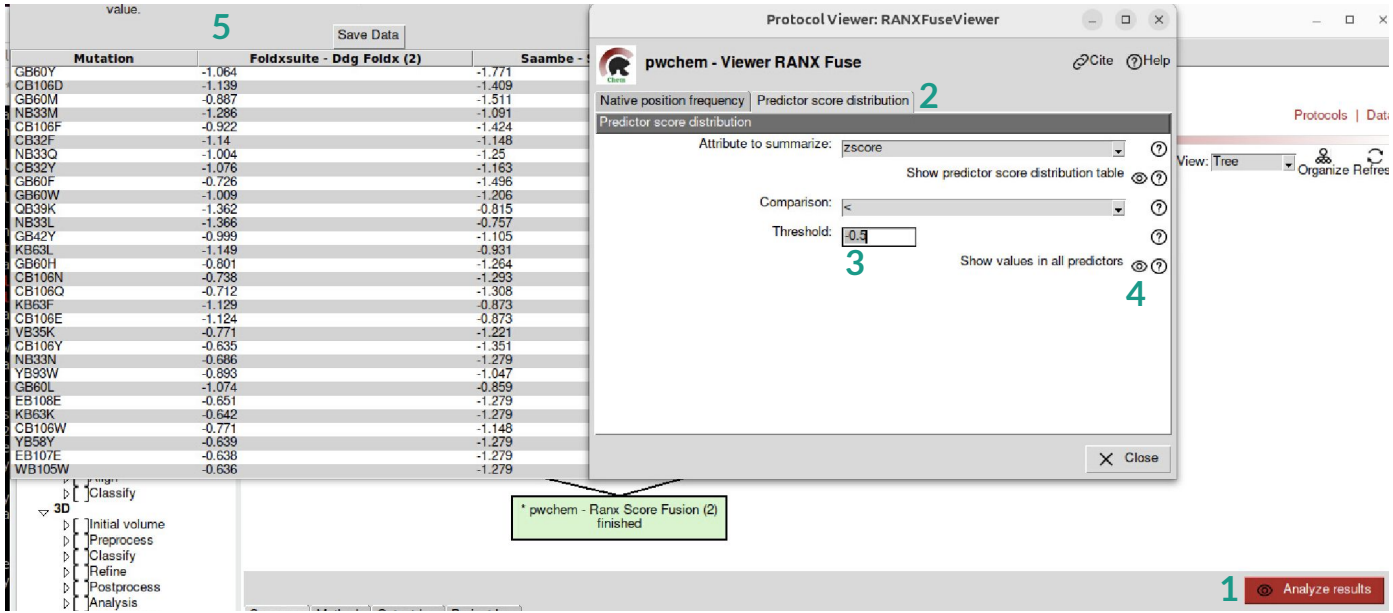
FoldX



# Rank mutations by z-score

FoldX

SAAMBE-3D



# Play around

Play around in the protein–protein interaction project, or create a new project for your complex of interest.

## Proposed activities

- Which protocol would you use to find interactions by bond type?
- Filter the set of mutations on Flex ddG: **z-score less than −1.5.** What can you conclude about these residues?
- Compare both lists: **which residues are flagged by more than one tool?**